

SECTION 1. IDENTIFICATION OF THE CHEMICAL AND COMPANY

Product name : SMARTPOWER DISHMACHINE DETERGENT HEAVY DUTY

Other means of identification : Not applicable

Recommended use : Machine Warewashing Detergent

Product dilution information : 0,1 % - 0,25 %

Company : Chile ECOLAB S.A.
Avenida Pedro de Valdivia 3801
Santiago, Santiago Chile
(2)-22413300, Telephone: (2)-22381603, Telephone: SAC: 600 241 6600
sac.chile@ecolab.com

Emergency telephone : (+56-2) 2635-3600 (CITUC) EMERGENCY PHONE

Toxicological Emergency Phone : CITUC (+56-2) 2635-3800 (24 hours) Toxicological Emergencies

SECTION 2. HAZARDS IDENTIFICATION

Classification according to NCh 382 : Not classified according to NCh 382

Distintivo según NCh 2190 : Not classified according to NCh 382

GHS Classification
Product AS SOLD

Acute toxicity (Oral) : Category 5
Eye irritation : Category 2A

Product AT USE DILUTION

Not a hazardous substance or mixture according to the Globally Harmonized System (GHS).

GHS label elements
Product AS SOLD

Hazard pictograms :



Signal Word : Warning

Hazard Statements : May be harmful if swallowed.
Causes serious eye irritation.

Precautionary Statements : **Prevention:**
Wash skin thoroughly after handling. Wear eye protection/ face protection.
Response:

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SMARTPOWER DISHMACHINE DETERGENT HEAVY DUTY

IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. Call a POISON CENTER/ doctor if you feel unwell. If eye irritation persists: Get medical advice/ attention.

Product AT USE DILUTION

Precautionary Statements : **Prevention:**
Wash hands thoroughly after handling.
Response:
Get medical advice/ attention if you feel unwell.
Storage:
Store in accordance with local regulations.

Product AS SOLD

Other hazards : None known.

SECTION 3. COMPOSITION/INFORMATION OF INGREDIENTS

Product AS SOLD

Pure substance/mixture : Mixture

Chemical name	CAS-No.	Concentration (%)
Sodium Carbonate	497-19-8	60 - 100
alanine, n,n-bis(carboxymethyl)-, trisodium salt	164462-16-2	5 - 10
sodium hydrogencarbonate	144-55-8	1 - 5
Alcohol alkoxylate		1 - 5
subtilisin	9014-01-1	0,1 - 1

Product AT USE DILUTION

No hazardous ingredients

SECTION 4. FIRST AID MEASURES

Product AS SOLD

In case of eye contact : Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Remove contact lenses, if present and easy to do. Continue rinsing. Get medical attention.

In case of skin contact : Rinse with plenty of water.

If swallowed : Rinse mouth. Get medical attention if symptoms occur.

If inhaled : Get medical attention if symptoms occur.

Protection of first-aiders : If potential for exposure exists refer to Section 8 for specific personal protective equipment.

Notes to physician : Treat symptomatically.

Product AT USE DILUTION

In case of eye contact : Rinse with plenty of water.

In case of skin contact : Rinse with plenty of water.

If swallowed : Rinse mouth. Get medical attention if symptoms occur.

If inhaled : Get medical attention if symptoms occur.

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SMARTPOWER DISHMACHINE DETERGENT HEAVY DUTY

Product AS SOLD

Expected acute effects

Eyes	: Causes serious eye irritation.
Skin	: Health injuries are not known or expected under normal use.
Ingestion	: May be harmful if swallowed.
Inhalation	: Health injuries are not known or expected under normal use.

Expected chronic effects

Chronic Exposure	: Health injuries are not known or expected under normal use.
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Product AT USE DILUTION

Expected acute effects

Eyes	: Health injuries are not known or expected under normal use.
Skin	: Health injuries are not known or expected under normal use.
Ingestion	: Health injuries are not known or expected under normal use.
Inhalation	: Health injuries are not known or expected under normal use.

Expected chronic effects

Chronic Exposure	: Health injuries are not known or expected under normal use.
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Most important symptoms / effects

Product AS SOLD

Eye contact	: Redness, Pain, Irritation
Skin contact	: No symptoms known or expected.
Ingestion	: No symptoms known or expected.
Inhalation	: No symptoms known or expected.

Product AT USE DILUTION

Eye contact	: No symptoms known or expected.
Skin contact	: No symptoms known or expected.
Ingestion	: No symptoms known or expected.
Inhalation	: No symptoms known or expected.

SECTION 5. FIREFIGHTING MEASURES

Product AS SOLD

Extinguishing Agents	: Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.
Unsuitable extinguishing agents	: None known.
Specific hazards during fire fighting	: Not flammable or combustible.

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SMARTPOWER DISHMACHINE DETERGENT HEAVY DUTY

Products generated in combustion and thermal degradation	: Decomposition products may include the following materials: Carbon oxides Nitrogen oxides (NO _x) Sulfur oxides Oxides of phosphorus
Precautions for Emergency Personnel and / or Firefighters	: Use personal protective equipment.
Specific extinguishing methods	: Fire residues and contaminated fire extinguishing water must be disposed of in accordance with local regulations.

SECTION 6. MEASURES IN CASE OF ACCIDENTAL SPILLAGE

Product AS SOLD

Personal precautions, protective equipment and emergency procedures	: Ensure clean-up is conducted by trained personnel only. Refer to protective measures listed.
Environmental precautions	: Do not allow contact with soil, surface or ground water.
Methods and materials for containment and cleaning up	: Sweep up and shovel into suitable containers for disposal.

Product AT USE DILUTION

Personal precautions, protective equipment and emergency procedures	: Refer to protective measures listed.
Environmental precautions	: No special environmental precautions required.
Methods and materials for containment and cleaning up	: Stop leak if safe to do so. Contain spillage, and then collect with non-combustible absorbent material, (e.g. sand, earth, diatomaceous earth, vermiculite) and place in container for disposal according to local / national regulations (see section 13). Flush away traces with water. For large spills, dike spilled material or otherwise contain material to ensure runoff does not reach a waterway.

SECTION 7. HANDLING AND STORAGE

Product AS SOLD

Handling

Advice on safe handling	: Do not ingest. Avoid contact with skin and eyes. Wash hands thoroughly after handling. In case of mechanical malfunction, or if in contact with unknown dilution of product, wear full Personal Protective Equipment (PPE).
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Storage

Conditions for safe storage	: Keep out of reach of children. Store in suitable labeled containers.
Recommended and unsuitable packaging by supplier	: Use high density plastic containers.
Storage temperature	: 0 °C to 50 °C

Product AT USE DILUTION

SAFETY DATA SHEET

SMARTPOWER DISHMACHINE DETERGENT HEAVY DUTY

Handling

Advice on safe handling : Wash hands after handling. In case of mechanical malfunction, or if in contact with unknown dilution of product, wear full Personal Protective Equipment (PPE). For personal protection see section 8.

Storage

Conditions for safe storage : Keep out of reach of children. Store in suitable labeled containers.

Recommended and unsuitable packaging by supplier : Use high density plastic containers.

SECTION 8. EXPOSURE CONTROLS/PERSONAL PROTECTION

Product AS SOLD

Ingredients with workplace control parameters

Components	CAS-No.	Form of exposure	Permissible concentration	Basis
subtilisin	9014-01-1	CMP-C	0,00006 mg/m ³ (crystalline active enzyme)	AR OEL
subtilisin	9014-01-1	CEIL	0,00006 mg/m ³	PE OEL
subtilisin	9014-01-1	VLE-P	0,00006 mg/m ³ (100% pure crystalline enzymes)	NOM-010-STPS-2014
subtilisin	9014-01-1	C	0,00006 mg/m ³ (crystalline active enzyme)	ACGIH
		STEL	0,00006 mg/m ³	NIOSH REL

Engineering measures : Good general ventilation should be sufficient to control worker exposure to airborne contaminants.

Personal protective equipment

Eye protection : Safety glasses with side-shields

Hand protection : No special protective equipment required.

Skin protection : No special protective equipment required.

Respiratory protection : No personal respiratory protective equipment normally required.

Hygiene measures : Handle in accordance with good industrial hygiene and safety practice. Wash face, hands and any exposed skin thoroughly after handling.

Product AT USE DILUTION

Engineering measures : Good general ventilation should be sufficient to control worker exposure to airborne contaminants.

Personal protective equipment

Eye protection : No special protective equipment required.

Hand protection : No special protective equipment required.

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Skin protection : No special protective equipment required.

Respiratory protection : No personal respiratory protective equipment normally required.

SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES

	Product AS SOLD	Product AT USE DILUTION
Appearance	: Extruded solid.	liquid
Color	: opaque, off-white	colorless
Odor	: odorless	odorless
pH	: 10,3 - 11,3, (1 %)	9,5 - 11,0
Flash point	: Not applicable	
Odor Threshold	: No data available	
Melting point/freezing point	: No data available	
Initial boiling point and boiling range	: 100 °C	
Evaporation rate	: No data available	
Flammability (solid, gas)	: Not applicable	
Upper explosion limit	: No data available	
Lower explosion limit	: No data available	
Vapor pressure	: No data available	
Relative vapor density	: No data available	
Relative density	: No data available	
Water solubility	: soluble	
Solubility in other solvents	: No data available	
Partition coefficient: n-octanol/water	: No data available	
Autoignition temperature	: No data available	
Thermal decomposition	: No data available	
Viscosity, kinematic	: No data available	
Explosive properties	: No data available	
Oxidizing properties	: No data available	
Molecular weight	: No data available	
VOC	: No data available	

SECTION 10. STABILITY AND REACTIVITY

Product AS SOLD

Reactivity : No dangerous reaction known under conditions of normal use.

Chemical stability : Stable under normal conditions.

Possibility of hazardous reactions : No dangerous reaction known under conditions of normal use.

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Conditions to avoid	: None known.
Incompatible materials	: Acids
Hazardous decomposition products	: In case of fire hazardous decomposition products may be produced such as: Carbon oxides Nitrogen oxides (NOx) Sulfur oxides Oxides of phosphorus

SECTION 11. TOXICOLOGICAL INFORMATION

Information on likely routes of exposure : Eye contact, Skin contact

Toxicity

Product AS SOLD Product

Acute oral toxicity	: Acute toxicity estimate : 2.914 mg/kg
Acute inhalation toxicity	: No data available
Acute dermal toxicity	: Acute toxicity estimate : > 5.000 mg/kg
Skin corrosion/irritation	: No data available
Serious eye damage/eye irritation	: No data available
Respiratory or skin sensitization	: No data available
Carcinogenicity	: No data available
Reproductive effects	: No data available
Germ cell mutagenicity	: No data available
Teratogenicity	: No data available
STOT-single exposure	: No data available
STOT-repeated exposure	: No data available
Aspiration toxicity	: No data available

SECTION 12. ECOLOGICAL INFORMATION

Product AS SOLD Ecotoxicity

Environmental Effects : This product has no known ecotoxicological effects.

Product

Toxicity to fish	: No data available
Toxicity to daphnia and other aquatic invertebrates	: No data available
Toxicity to algae	: No data available

Components

Toxicity to fish	: Sodium Carbonate
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96 h LC50 *Lepomis macrochirus* (Bluegill sunfish): 300 mg/l

alanine, n,n-bis(carboxymethyl)-, trisodium salt
96 h LC50: > 200 mg/l

Alcohol alkoxylate
96 h LC50 *Pimephales promelas*: 11,3 mg/l

Components

Toxicity to daphnia and other aquatic invertebrates : Sodium Carbonate
48 h EC50 *Ceriodaphnia* (water flea): 213,5 mg/l

sodium hydrogencarbonate
48 h EC50 *Daphnia*: 41.000 mg/l

Alcohol alkoxylate
48 h EC50 *Daphnia magna* (Water flea): > 100 mg/l

subtilisin
48 h EC50: 1,4 mg/l

Components

Toxicity to algae : Alcohol alkoxylate
72 h EC50 *Pseudokirchneriella subcapitata* (green algae): 67,9 mg/l

Persistence and degradability

Product AS SOLD

No data available

Product AT USE DILUTION

Readily biodegradable.

Bioaccumulative potential

No data available

Mobility in soil

No data available

Other adverse effects

No data available

SECTION 13. INFORMATION OF DISPOSAL

Product AS SOLD

Disposal of contaminated packaging : In accordance with what is described by Supreme Decree No. 148, the packaging of the product is considered hazardous waste and must be disposed of through companies authorized to receive and / or treat such waste, which must be issued and final disposal certificate waste.

Product AT USE DILUTION

SECTION 14. TRANSPORT INFORMATION

Product AS SOLD

SAFETY DATA SHEET

SMARTPOWER DISHMACHINE DETERGENT HEAVY DUTY

The shipper/consignor/sender is responsible to ensure that the packaging, labeling, and markings are in compliance with the selected mode of transport.

Land transport

Official transport classification of the United Nations

Not dangerous goods

Air transport (IATA)

Official transport classification of the United Nations

Contact Regulatory for air freight eligibility

Sea transport (IMDG/IMO)

Official transport classification of the United Nations

Not dangerous goods

SECTION 15. REGULATORY INFORMATION

Registrations and Certifications

Chile: Our HDS meets Chilean Normative: NCH 2245: 2015 (Safety Data Sheet for chemical products - Content and order of sections)

The receiver should pay attention to the possible existence of local regulations.

NCh 1411: Prevention of risks, IV identification of risks of materials

NCh 2190: Transportation of Hazardous Substances - Pictograms for hazard identification

NCh 382: Dangerous Good - Classification

D.S. No. 594: Minimum basic conditions in the workplace

D.S. No. 148: Hazardous waste disposal

D.S. No. 72: Mining Safety Regulations

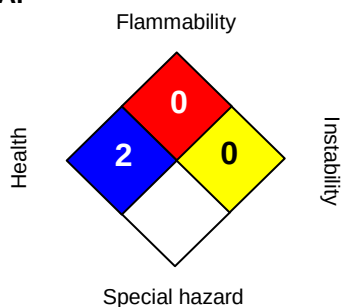
D. S. No. 43: reports on the storage of hazardous substances

D. S. No. 40: Reporting on exposure hazards

SECTION 16. OTHER INFORMATION

Product AS SOLD

NFPA:



HMIS III:

HEALTH	2
FLAMMABILITY	0
PHYSICAL HAZARD	0

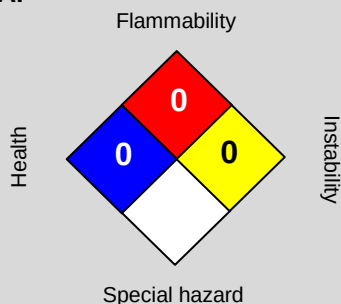
0 = not significant, 1 = Slight,
2 = Moderate, 3 = High
4 = Extreme, * = Chronic

Product AT USE DILUTION

SAFETY DATA SHEET

SMARTPOWER DISHMACHINE DETERGENT HEAVY DUTY

NFPA:



HMIS III:

HEALTH	0
FLAMMABILITY	0
PHYSICAL HAZARD	0

0 = not significant, 1 = Slight,
2 = Moderate, 3 = High
4 = Extreme, * = Chronic

Issuing date : 22.06.2020
Version : 1.0
Prepared by : Regulatory Affairs

While there is no change in the formula or hazard classifications, this SDS remains valid.

REVISED INFORMATION: Significant changes to regulatory or health information for this revision is indicated by a bar in the left-hand margin of the SDS.

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.